Literature-Implied Status for Question CIC

Run `fa_banach_001`, agent `agent_lane_18`

2026-07-03

Status

Literature-implied answer, human verification recommended.

The source problem is Question CIC in Cwikel and Rochberg, *Nigel Kalton and complex interpolation of compact operators*, arXiv:1410.4527. A recent preprint of Pustylnik, *A new approach to interpolation of compact linear operators*, arXiv:2605.00530, appears to imply an affirmative answer.

Source Question

For Banach couples (X_0, X_1) and (Y_0, Y_1), Cwikel and Rochberg write

$$(X_0, X_1) \blacktriangleright (Y_0, Y_1)$$

when every linear operator $T : X_0 + X_1 \to Y_0 + Y_1$ that is compact $X_0 \to Y_0$ and bounded $X_1 \to Y_1$ is compact

$$[X_0, X_1]_\theta \to [Y_0, Y_1]_\theta$$

for every $0 < \theta < 1$. Their Question CIC asks whether this property holds for all Banach couples.

Supporting Paper

The relevant source passages in arXiv:2605.00530 are:

- lines 270–284: the paper distinguishes two-sided compactness from one-sided compactness and says one-sided compactness requires condition (2);
- lines 336–344: condition (2) is the estimate

$$\|T\|_{A\to B} \leq W(\|T\|_{A_0\to B_0}, \|T\|_{A_1\to B_1}),$$

  with $W(\alpha, \beta) \to 0$ as $\beta \to 0$ for bounded α, and the complex interpolation functor is explicitly covered with $W(\alpha, \beta) = \alpha^{1-\theta}\beta^\theta$;
- lines 346–349: the Main Theorem concludes compactness of the interpolated embedding under these hypotheses;
- lines 598–610: the paper passes from embedding operators to arbitrary linear operators via image spaces.

Identification

Question CIC is the one-sided compactness problem for the complex interpolation functor. Pustylnik's one-sided theorem applies once condition (2) holds, and the paper explicitly states that condition (2) holds for the complex functor. The final image-space reduction then converts the embedding theorem into the corresponding statement for arbitrary operators. Thus, if arXiv:2605.00530 is correct as stated, it gives a full affirmative answer to Question CIC.

Caveat

This is not presented in arXiv:2605.00530 as an explicit answer to Cwikel–Rochberg's Question CIC, and the supporting paper is a recent broad preprint about a famously difficult problem. This packet should therefore be treated as a literature-implied claimed answer pending human review of the proof.

References

[1] M. Cwikel and R. Rochberg, *Nigel Kalton and complex interpolation of compact operators*, arXiv:1410.4527, 2014.

[2] E. Pustylnik, *A new approach to interpolation of compact linear operators*, arXiv:2605.00530, 2026.